

Bitwise differential trail through **TWINE** found by SAT

CiVerLy

July 24, 2026

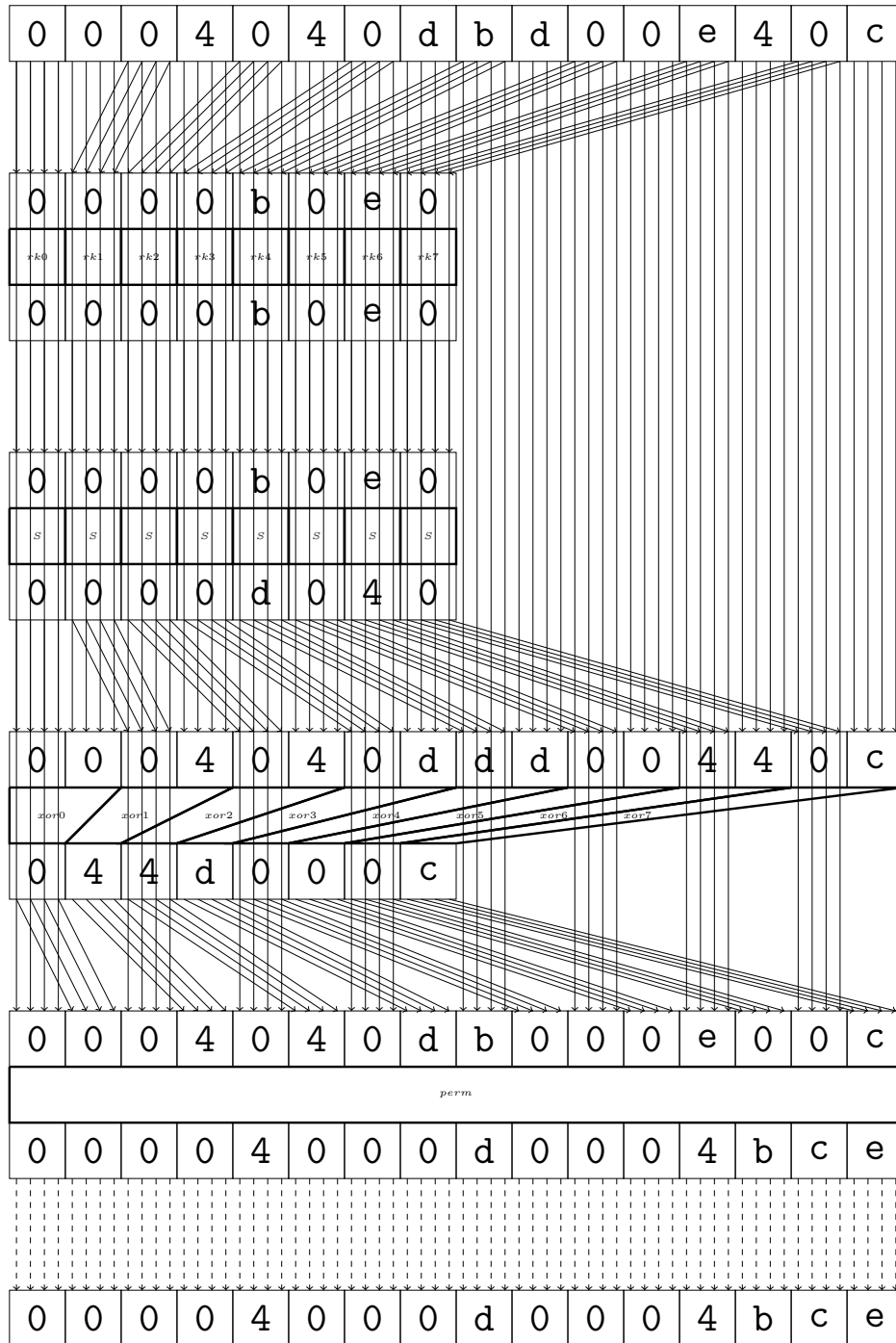
1 TWINE

Maximal differential probability: 2^{-46}

0	0	0	4	0	4	0	d	b	d	0	0	e	4	0	c
0	0	0	4	0	4	0	d	b	d	0	0	e	4	0	c
round															
0	0	0	0	4	0	0	0	d	0	0	0	4	b	c	e
0	0	0	0	4	0	0	0	d	0	0	0	4	b	c	e
round															
0	0	0	0	0	0	e	4	0	0	0	c	b	d	0	4
0	0	0	0	0	0	e	4	0	0	0	c	b	d	0	4
round															
0	0	c	e	0	0	0	0	0	0	0	0	0	4	b	
0	0	c	e	0	0	0	0	0	0	0	0	0	4	b	
round															
0	c	0	0	0	0	0	0	0	0	0	4	0	0	0	0
0	c	0	0	0	0	0	0	0	0	0	4	0	0	0	0
round															
c	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0
c	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0
round															
e	4	0	0	b	c	0	0	0	0	0	0	0	0	0	0
e	4	0	0	b	c	0	0	0	0	0	0	0	0	0	0
round															
0	0	0	0	0	e	0	b	0	0	0	0	0	0	0	0
0	0	0	0	0	e	0	b	0	0	0	0	0	0	0	0
round															
0	0	0	0	0	0	0	b	0	0	0	e	0	0	0	0
0	0	0	0	0	0	0	b	0	0	0	e	0	0	0	0
round															
0	0	0	0	0	0	d	0	0	0	4	0	0	b	0	e
0	0	0	0	0	0	d	0	0	0	4	0	0	b	0	e
round															
0	0	b	d	0	0	0	0	1	4	b	0	0	0	e	0
0	0	b	d	0	0	0	0	1	4	b	0	0	0	e	0
round															
0	0	b	0	0	0	0	0	1	8	b	d	0	0	e	4
0	0	b	0	0	0	0	0	1	8	b	d	0	0	e	4

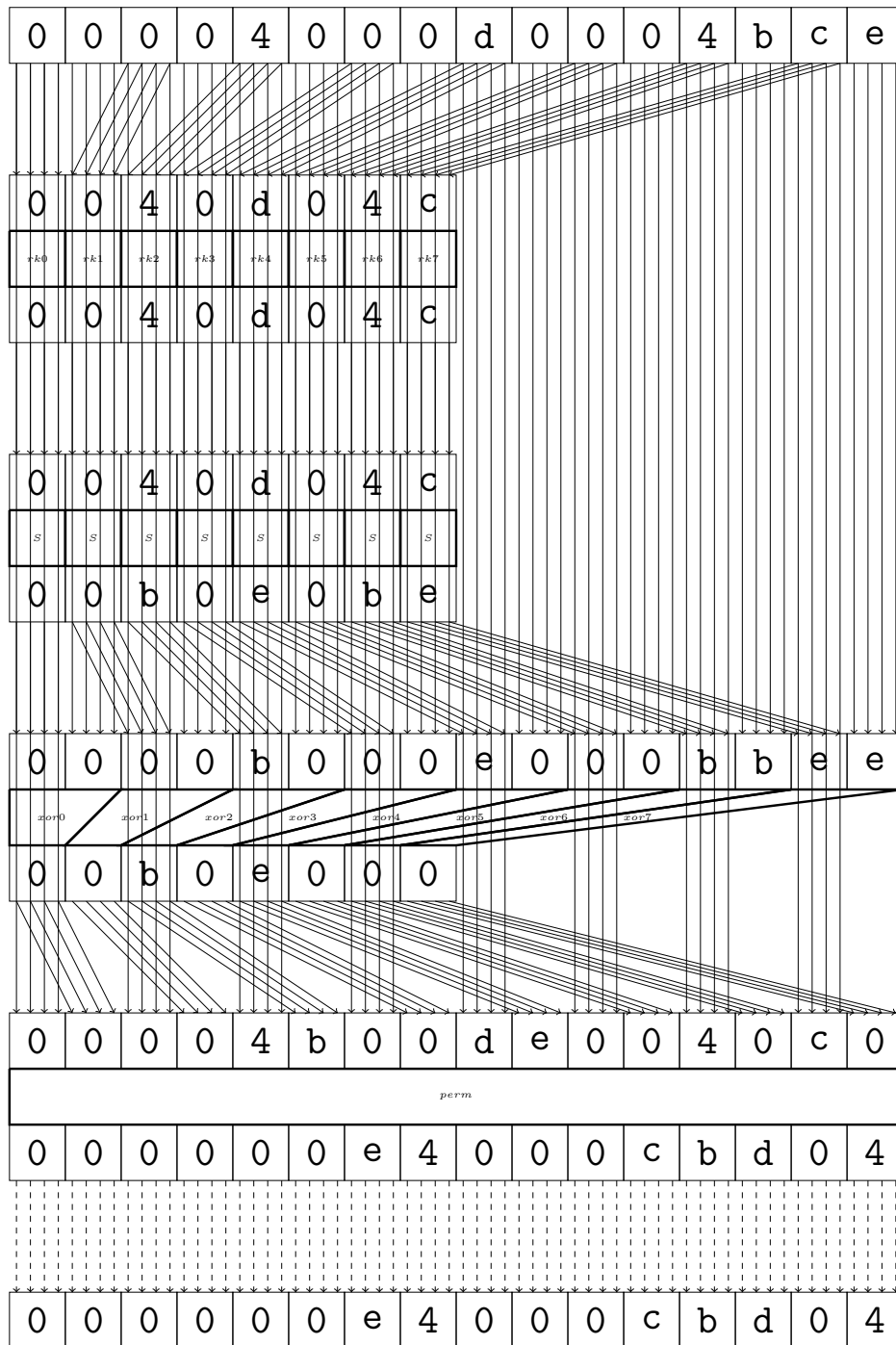
2 round

Maximal differential probability: $2^{-4.0}$



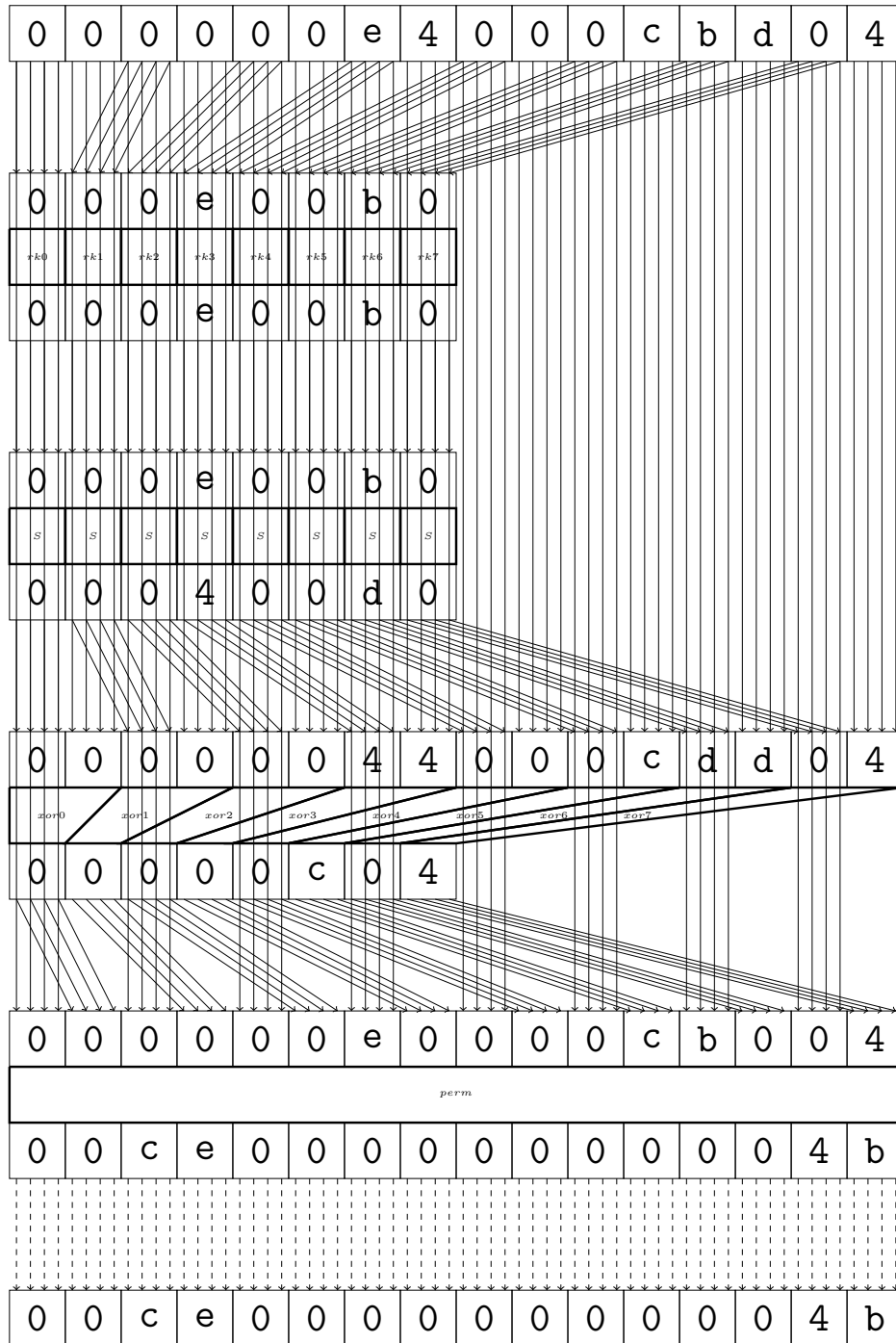
3 round

Maximal differential probability: $2^{-9.0}$



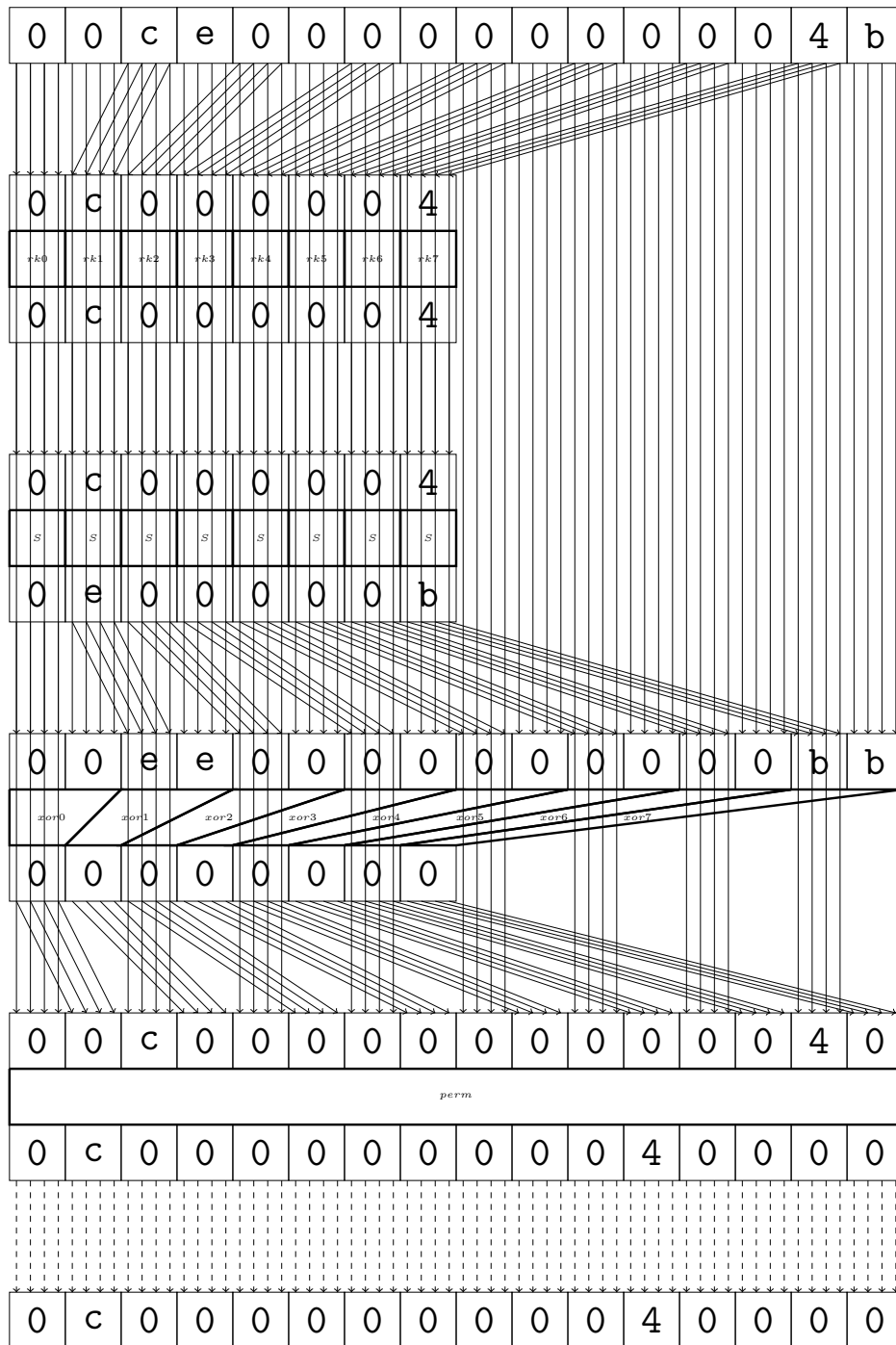
4 round

Maximal differential probability: $2^{-4.0}$



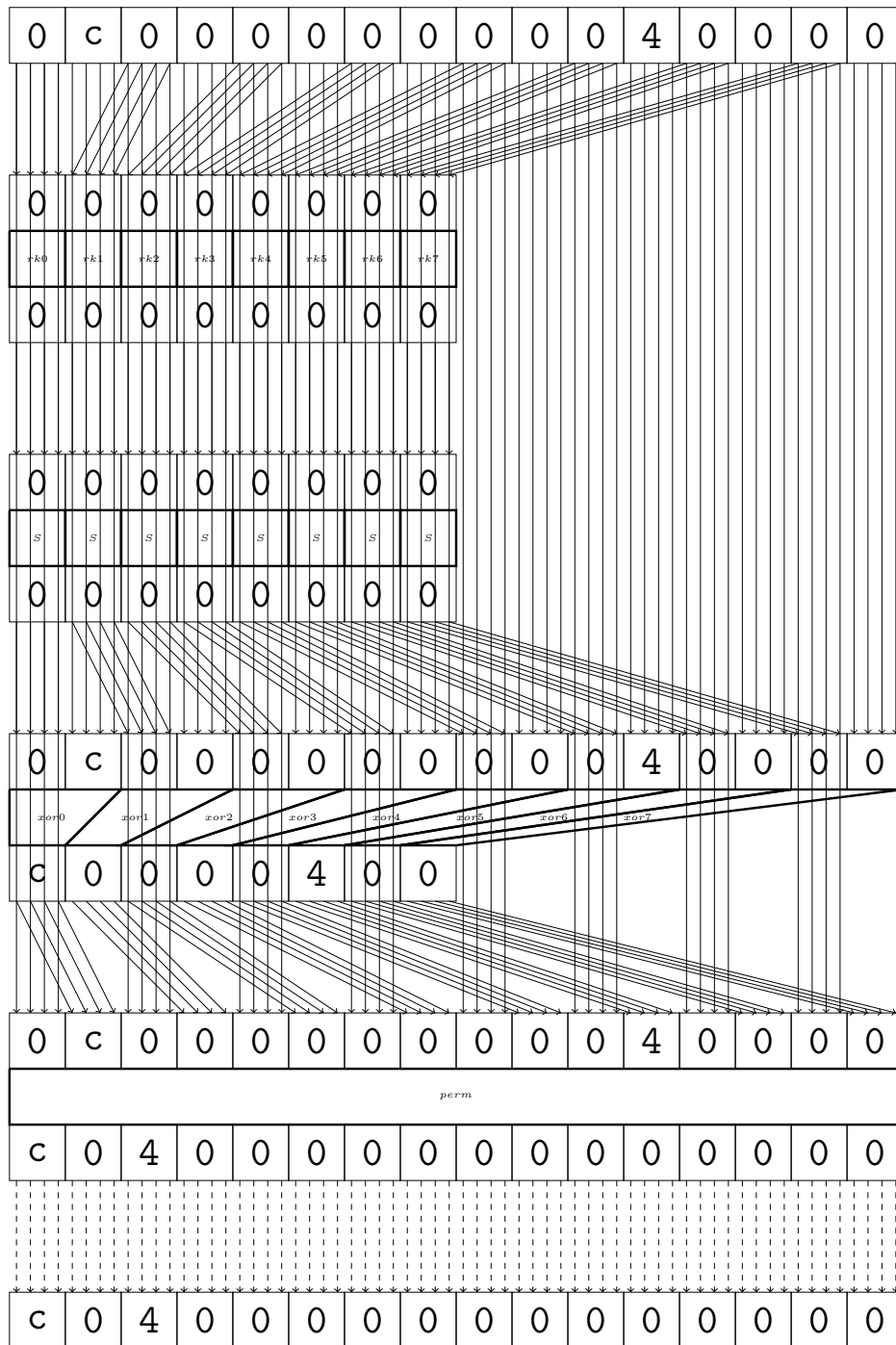
5 round

Maximal differential probability: $2^{-4.0}$

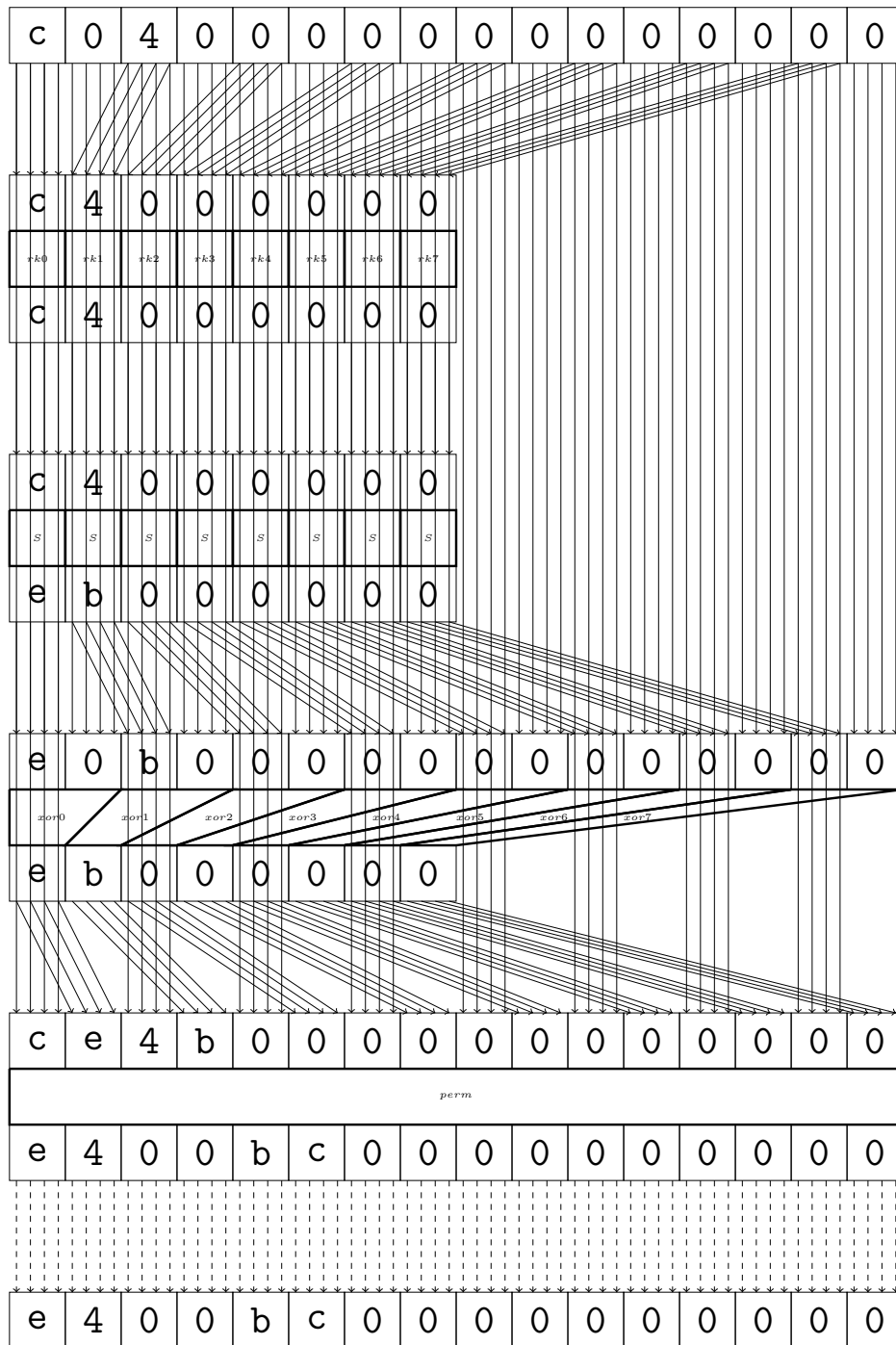


6 round

Maximal differential probability: $2^{-0.0}$

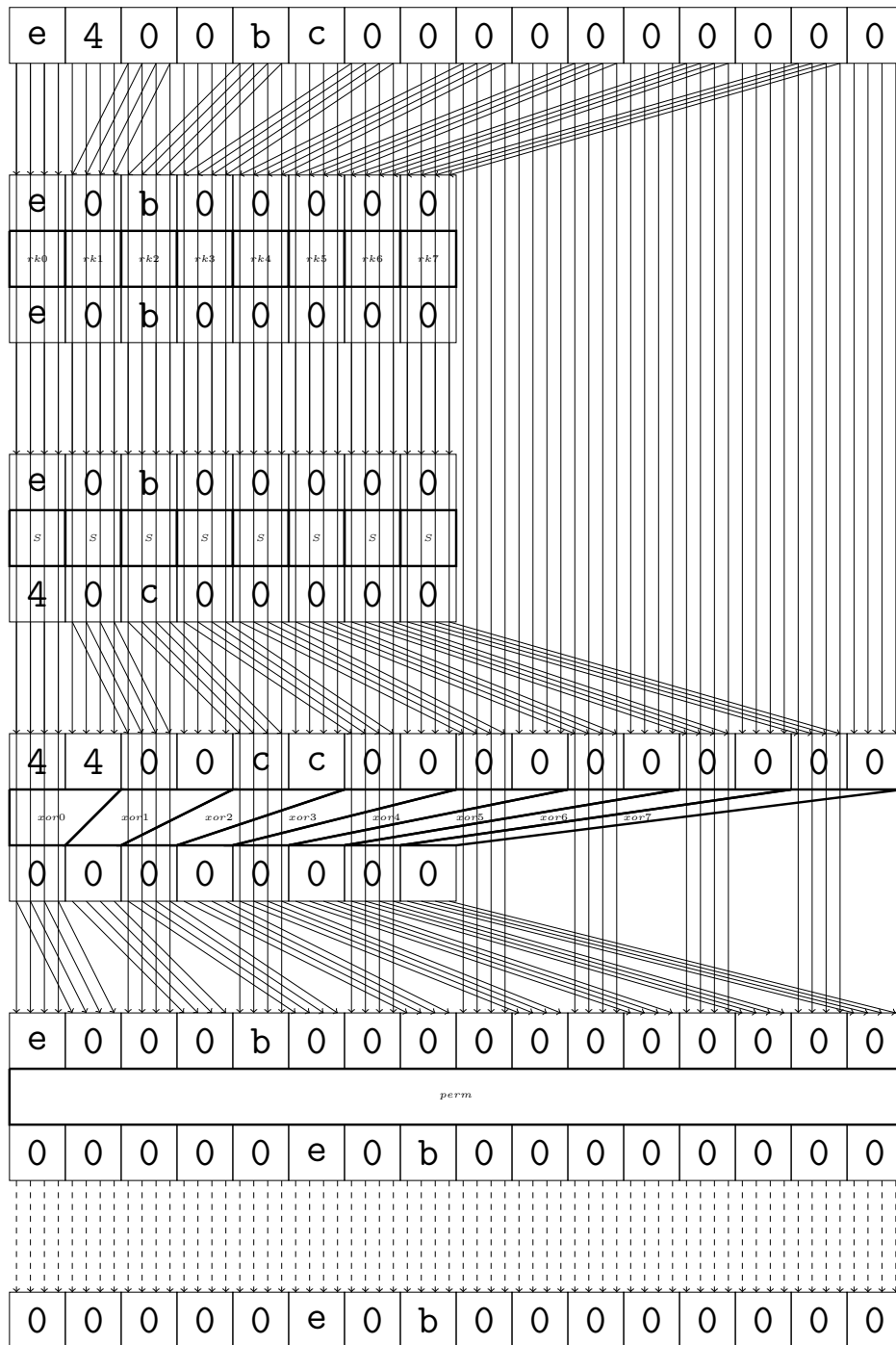


7 round

Maximal differential probability: $2^{-4.0}$ 

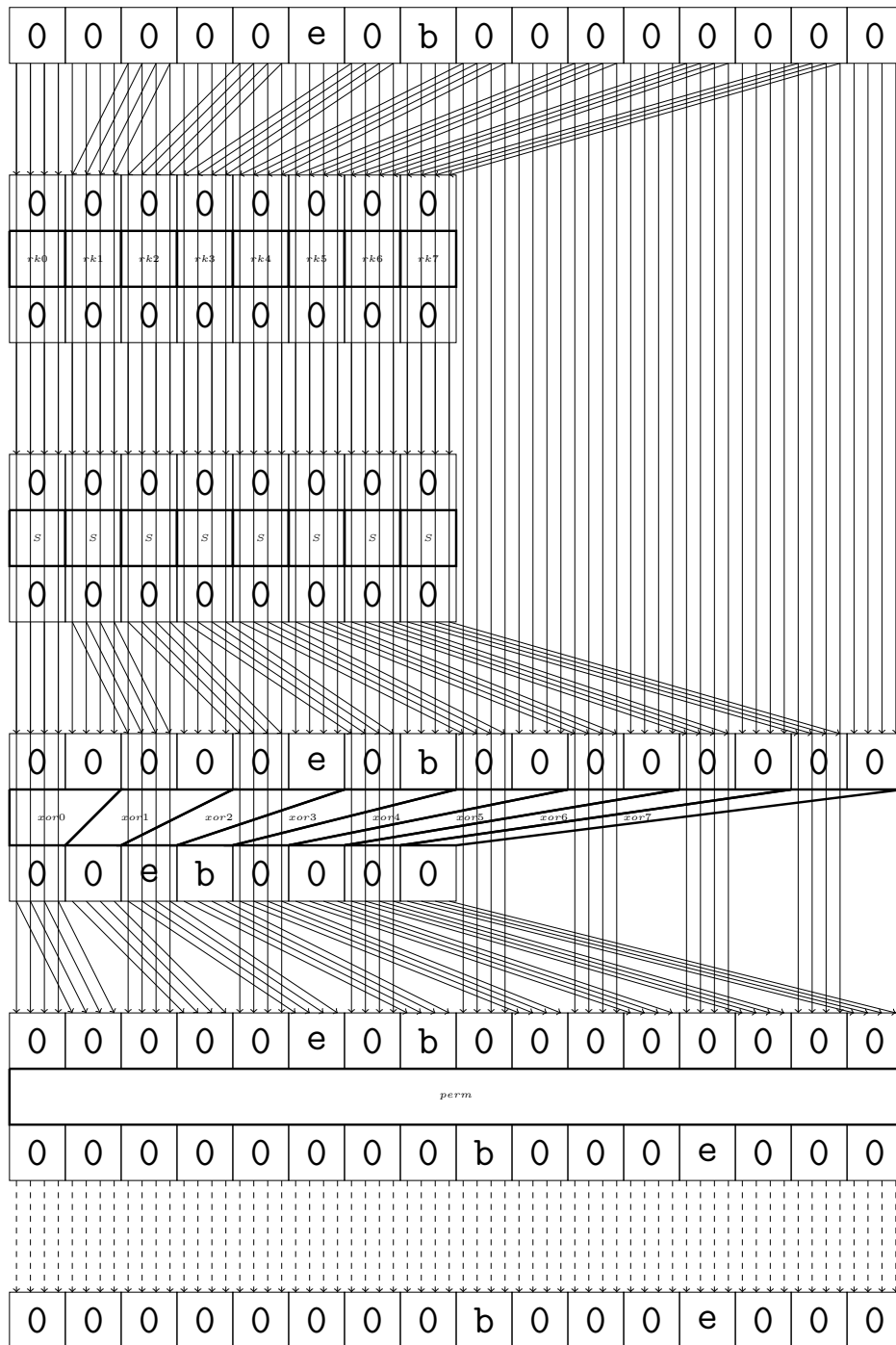
8 round

Maximal differential probability: $2^{-5.0}$



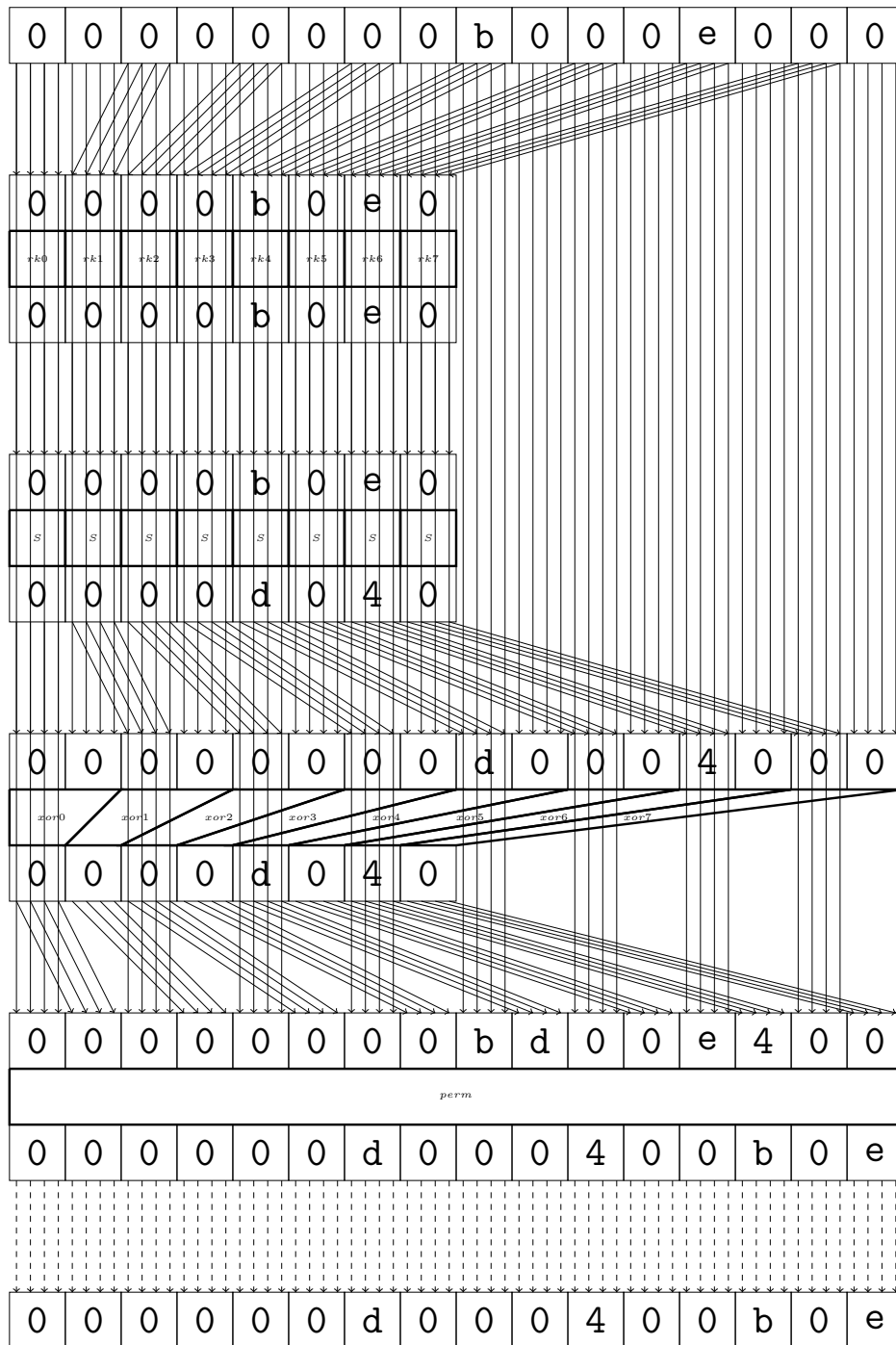
9 round

Maximal differential probability: $2^{-0.0}$



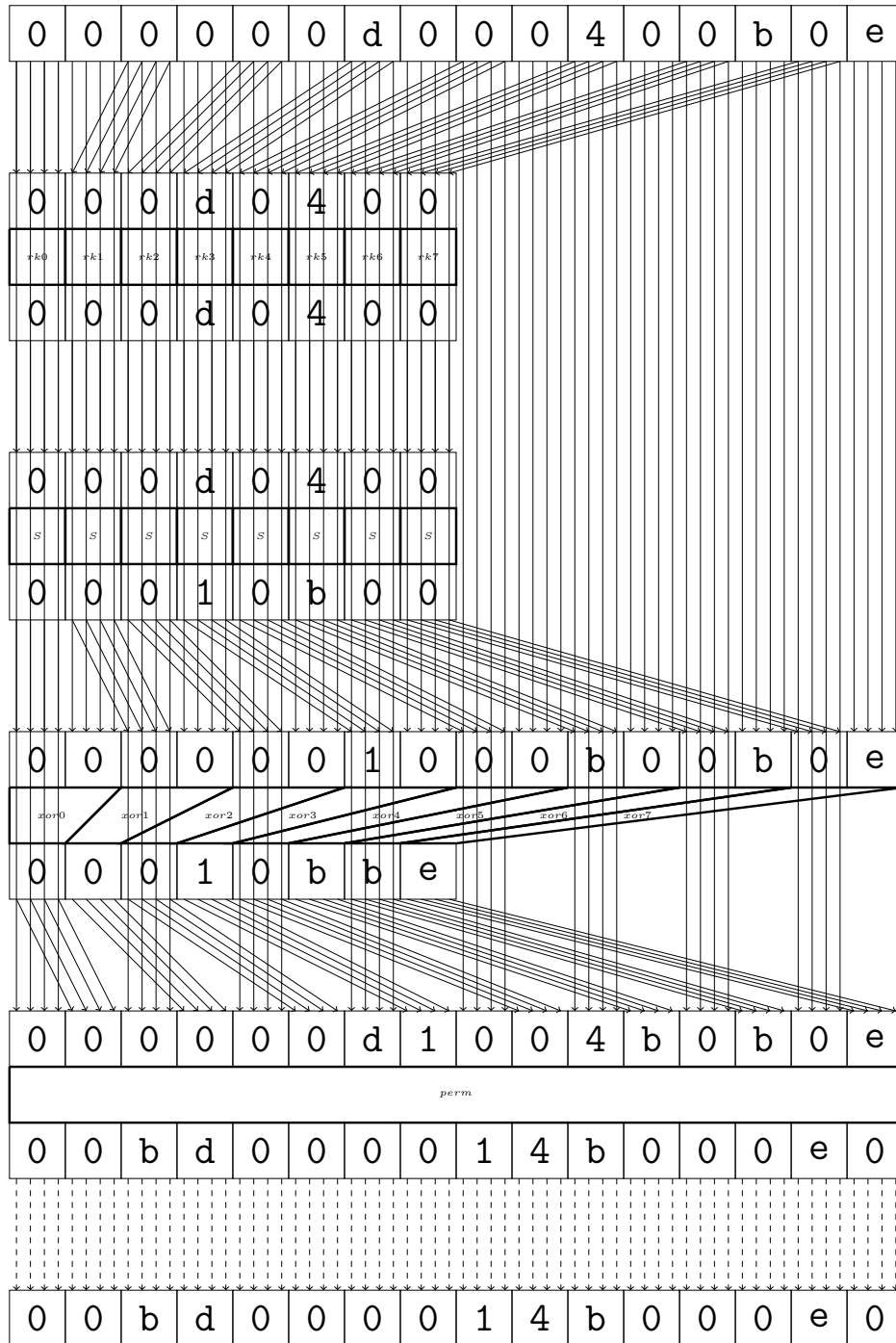
10 round

Maximal differential probability: $2^{-4.0}$



11 round

Maximal differential probability: $2^{-4.0}$



12 final

Maximal differential probability: $2^{-8.0}$ 